

Adnan Armouti

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Interests: Physical AI, RF Spatial Intelligence, Differentiable Rendering, Inverse Problems, Radar

Education

Ph.D. Candidate in Computer Science , Cornell University	Aug 2026–Present
<i>Advisor:</i> Rajalakshmi Nandakumar	
M.S., Computer Science , Cornell University	Aug 2023–May 2026
M.S., Electrical and Computer Engineering , UCLA	Sep 2021–Jun 2023
B.S., Electrical Engineering , UCLA	Sep 2017–Jun 2021
B.A., Economics , UCLA	Sep 2017–Jun 2021

Current Research

Towards RF Spatial Intelligence: Physics-Based Differentiable Rendering for Inverse Problems in Millimeter-Wave Radar

Committee: Rajalakshmi Nandakumar (Chair), Andrew Owens, Steve Marschner, Alyosha Molnar

- I develop explicit, physics-based differentiable renderers for mmWave radar that fit real captures via material and surface normal estimation [P6, P5].
- Explicit modeling of independent scene and sensor representations enables scene- and sensor-level editing to solve inverse problems such as 2D-to-3D lifting [P5], super-resolution [P5], and novel view synthesis [P6].

Publications

Selected Papers

* equal contribution

- [P6] **Adnan Armouti**, Yixuan Gao, Rajalakshmi Nandakumar. “3D Point Splatting for mmWave Radar Novel View Synthesis.” *NeurIPS 2026*.
 - [P5] **Adnan Armouti**, Yixuan Gao, Rajalakshmi Nandakumar. “mmIR: Frequency-Space Inverse Rendering for 3D Millimeter-Wave Radar ADC Synthesis.” *ECCV 2026*.
 - [P4] Alexander Vilesov*, Pradyumna Chari*, **Adnan Armouti***, Anirudh Bindiganavale Harish, Kimaya Kulkarni, Ananya Deoghare, Laleh Jalilian, Achuta Kadambi. “Blending Camera and 77 GHz Radar Sensing for Equitable, Robust Plethysmography.” *ACM Transactions on Graphics (SIGGRAPH 2022)*.
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- [P3] Tanvir Ahmed, Yixuan Gao, **Adnan Armouti**, Rajalakshmi Nandakumar. “mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption.” *MobiCom 2026*.
 - [P2] Pradyumna Chari*, Anirudh Bindiganavale Harish*, **Adnan Armouti**, Alexander Vilesov, Sanjit Sarda, Laleh Jalilian, Achuta Kadambi. “Implicit Neural Models to Extract Heart Rate from Video.” *ECCV 2024*.
 - [P1] Kai Del Regno, Alexander Vilesov, **Adnan Armouti**, Anirudh Bindiganavale Harish, Selim Emir Can, Ashley Kita, Achuta Kadambi. “Thermal Imaging and Radar for Remote Sleep Monitoring of Breathing and Apnea.” *arXiv preprint (2024)*.

Patents

- [US2] Achuta Kadambi, Ashley Kita, Alexander Vilesov, Kai Del Regno, Selim Emir Can, Laleh Jalilian, Anirudh Bindiganavale Harish, **Adnan Armouti**. “Methods and Apparatus to Detect and Classify Forms of Sleep Apnea.” U.S. Provisional Application, 2024.
- [US1] Achuta Kadambi, Laleh Jalilian, Pradyumna Chari, Chinmay Talegaonkar, Doruk Karınca, Maxime Cannesson, Krish Kabra, Omid Salehi-Abari, Ashley Kita, **Adnan Armouti**. “[Systems and Methods for Measuring Vital Signs Using Multimodal Health Sensing Platforms](#).” U.S. Patent Application 2023/0233091, 2023.

Honors and Awards

NVIDIA Academic Grant	Jun 2026
UCLA ECE Outstanding Master of Science Student Award	Jun 2023
UCLA ECE Distinguished Master’s Thesis Research Award	May 2023
UCLA ECE VMG GSR Scholarship	Jan 2022
NSF REU Fellowship	Jun 2021
Intel SRC URP Scholarship	Jun 2020
UCLA ECE Dean’s Honors List	2020, 2021

Research Experience

Wireless Systems Lab, Cornell University, New York, NY Aug 2023–Present
PI: Rajalakshmi Nandakumar

- Built mmIR, an open-source differentiable FMCW radar inverse renderer with multi-bounce propagation, polarization, and diffraction; 0.914 RA correlation on 13 scenes vs. Sionna-RT’s 0.307 [P5].
- Built 3DPS, the first differentiable point renderer for radar novel view synthesis, producing ADC, range profiles, and range–azimuth maps from one model; 1.7–5.2 $\times$ over NVS baselines [P6].
- Contributed to mmFHE, the first mmWave sensing pipeline (signal processing and ML inference) run end to end under fully homomorphic encryption [P3].

Visual Machines Group, UCLA Sep 2020–Jun 2023
PI: Achuta Kadambi

- Led a successful \$1M DARPA grant proposal for contactless, equitable health sensing.
- Co-led camera–radar fusion for skin-tone-equitable remote plethysmography [P4].
- Co-implemented a neural implicit decomposition for heart-rate estimation on out-of-distribution subjects [P2], and contributed to thermal and radar sleep-apnea monitoring [P1].

Public Demonstrations

Tanvir Ahmed, Yixuan Gao, **Adnan Armouti**, Rajalakshmi Nandakumar. “[mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption](#).” *MobiCom Demos 2026* [P3].

Media Coverage

Forbes [P4]	Jan 2023
Daily Bruin [P4]	Sep 2022
UCLA Newsroom [P4]	Aug 2022

Open-Source Contributions

3DPS ([project page](#)). Differentiable point splatting for radar novel view synthesis [P6].
mmIR ([project page](#)). Differentiable FMCW radar inverse renderer [P5].
EquiPleth ([GitHub](#)). Camera–radar fusion for equitable plethysmography [P4].
FastImplicitPleth ([GitHub](#)). Implicit neural models for heart rate from video [P2].
NonContactApneaDetection ([GitHub](#)). Thermal and radar sleep-apnea detection [P1].

Theses

Fusing Light and Radio Signals for More Accurate and Equitable Blood Flow Estimation
M.S. Thesis, UCLA 2023
Committee: Achuta Kadambi (Chair), Jonathan Kao, Cho-Jui Hsieh, Bolei Zhou

Teaching

Teaching Assistant, Cornell Tech, Cornell University Aug 2023–Present

- CS 5785 “Applied Machine Learning,” Prof. Kyra Gan (Fall 2023, Fall 2024, Fall 2025, Fall 2026)
- CS 5788 “Introduction to Generative Models,” Prof. Andrew Owens (Spring 2026)
- INFO 5368 “Practical Applications in Machine Learning,” Prof. Angelique Taylor (Spring 2025, Spring 2026)

Mentorship

Guanyue (Thomas) Qian, <i>B.S. Student, NYU</i>	Jun 2026–Present
Kai Del Regno, <i>M.S. Student, UCLA</i>	Sep 2022–Jun 2023
Jianchong (Devin) Ma, <i>M.S. Student, Stanford University</i>	Sep 2022–Jun 2023
Zoe (Rui) Ma, <i>M.S. Student, Columbia University</i>	Sep 2022–Jun 2023

Referee Service

NeurIPS 2026–

Skills

- **AI / ML / Computer Vision:** PyTorch, JAX, TensorFlow
- **Computer Graphics:** Mitsuba 3, Dr.Jit, Slang, CUDA
- **Programming & Tools:** Python, C/C++, CUDA, MATLAB; Git, Linux, Docker, L^AT_EX

References

Prof. Rajalakshmi Nandakumar, Cornell University, rn283@cornell.edu
Prof. Andrew Owens, Cornell University, andrew.owens@cornell.edu
Prof. Kyra Gan, Cornell University, jg2424@cornell.edu
Prof. Achuta Kadambi, UCLA, achuta@ee.ucla.edu